



**MZANSI TECH CONTRACTORS**  
**PAYROLL SYSTEM**  
**COMPREHENSIVE TEST REPORT**  
**PM-04 Practical Module SDT621**  
**Date: 13 May 2026**  
**Version: 1.0 (Final)**  
**Tiaan Gerber 2024087G**

# Contents

|                                                                                                             |    |
|-------------------------------------------------------------------------------------------------------------|----|
| REPOSITORY OVERVIEW.....                                                                                    | 6  |
| EXECUTIVE SUMMARY.....                                                                                      | 7  |
| 1. TEST ENVIRONMENT.....                                                                                    | 7  |
| Software Environment .....                                                                                  | 7  |
| Test Data: The following five contractor records were used as test data throughout the testing process..... | 7  |
| 2. TESTING SCOPE .....                                                                                      | 10 |
| 3. TEST CASE EXECUTION RESULTS.....                                                                         | 10 |
| 3.2 Integration Testing Results.....                                                                        | 14 |
| 3.3 Validation Testing Results.....                                                                         | 15 |
| 3.4 System Testing Results.....                                                                             | 21 |
| 3.5 Regression Testing Results.....                                                                         | 26 |
| 4. DEFECT REPORT .....                                                                                      | 33 |
| 5. NON-FUNCTIONAL TESTING .....                                                                             | 34 |
| 6. MASTER TEST SUMMARY.....                                                                                 | 35 |
| 7. DEPLOYMENT READINESS DECISION .....                                                                      | 36 |

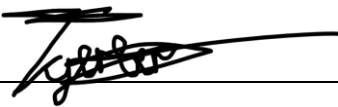




I, Tiaan Geber hereby

declare that the contents of this assignment **PM-04 Practical Module SDT621**  
is entirely my own work except for the following documents:

(List the documents and page numbers of work in this portfolio that were generated in a group)

Signature:  \_\_\_\_\_

## REPOSITORY OVERVIEW

All source code for the Mzansi Tech Contractors Payroll System has been uploaded to GitHub for verification purposes. The repository contains the complete Visual Studio solution including the main Windows Forms application project named PayrollApp, the console-based test suite project named PayrollTests, all forms, models, services, helper classes, and program files. The code compiles without errors and runs as documented in this testing report. The repository is structured to allow the examiner to easily navigate through the source code and verify that all functionality described in this document has been correctly implemented.

### REPOSITORY ACCESS

The repository can be accessed using the following link:

<https://github.com/tcoder2024/tcoder2024-MzansiTech-Payroll-System-PM04.git>

The repository is set to Public, allowing the examiner to access all files without requiring login credentials or access requests. The code in the repository matches the ZIP file submitted alongside this document, ensuring consistency between the online and offline submission materials.

*"tcoder2024" with your actual GitHub username*

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### REPOSITORY CONTENTS

The repository includes the complete Mzansi Tech Payroll System solution with the following structure:

#### **PayrollApp Project (Main Windows Forms Application):**

- The Forms folder contains the main application form FrmPayroll.cs which handles all user interface interactions including input validation, calculation triggering, result display, and history panel management. The designer file FrmPayroll.Designer.cs contains the layout configuration for the professional two-column interface with header, footer, input panel, results panel, and history grid.
- The Helpers folder contains ValidationHelper.cs which provides all input validation methods including name validation, hours range checking, dependent range validation, and safe numeric parsing.
- The Models folder contains PayrollRecord.cs which defines the data structure for each payroll record including contractor name, hours worked, dependents, all calculated values, and the calculation timestamp. This class also handles CSV serialization and deserialization.
- The Services folder contains two critical classes. CSVStorage.cs manages all file I/O operations including saving records to the payroll\_history.csv file located in the user Documents folder, loading existing records on application startup, and clearing history when requested. PayrollCalculator.cs contains all business logic including the CalculateGrossPay, CalculateUIF, CalculateMembershipFee, CalculatePAYE, CalculateNetPay, and CalculateAll methods. All calculation formulas are implemented exactly as specified in the assessment business rules.
- The Program.cs file serves as the application entry point and initializes the Windows Forms environment before launching the main FrmPayroll form.

#### **PayrollTests Project (Console Test Suite):**

- The Program.cs file contains the comprehensive test suite that reads all saved records from the main application's CSV storage, verifies each calculation independently, and displays results in professionally formatted ASCII tables with colour-coded PASS or FAIL status. The test suite executes twenty-five unit tests covering Gross Pay, UIF, Membership Fee, PAYE, and Net Pay calculations across all five test records. It also performs integration testing on the complete calculation chain, validation testing on all six business rules, system testing on the five-step application workflow, regression testing for cross-record consistency, and retesting verification for all corrected defects.

#### **Solution Files:**

- The PayrollSystem.sln solution file ties both projects together and allows the examiner to open the complete solution in Visual Studio with a single click.
- Both project files (PayrollApp.csproj and PayrollTests.csproj) are configured for .NET 8.0-windows framework with proper references and dependencies.

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## VERIFICATION INSTRUCTIONS

### Method One - Cloning the Repository:

Open a terminal or command prompt and execute the command `git clone https://github.com/tcoder2024/tcoder2024-MzansiTech-Payroll-System-PM04.git` to download a local copy of the repository. Navigate into the cloned folder and double-click the `PayrollSystem.sln` file to open the solution in Visual Studio. Press F5 to build and run the application. All functionality described in this testing report should be observable including proper input validation, accurate payroll calculations, CSV data persistence, history grid display, and the comprehensive test suite output.

### Method Two - Downloading as ZIP:

Navigate to the repository URL in any web browser. Click the green "Code" button and select "Download ZIP" from the dropdown menu. Extract the downloaded ZIP file to a convenient location on the examiner's computer. Open the extracted folder and double-click the `PayrollSystem.sln` file to launch the solution in Visual Studio. Press F5 to build and run the application.

### Expected Observable Functionality:

Once the application is running, the examiner should be able to enter contractor names, hours worked, and dependent counts. Clicking the Calculate Payroll button should display correct Gross Pay, UIF, Membership Fee, PAYE, and Net Pay values. The Net Pay amount should change colour based on the value green for high amounts, teal for medium amounts, gold for moderate amounts, and red for low amounts. Clicking the Save Record button should persist the record to the CSV file located in the user Documents folder under `PayrollSystem`. Clicking the View History button should expand a panel showing a `DataGridView` with all saved records including contractor names, hours, Gross Pay, Net Pay, and calculation dates. The Reset button should clear all input fields and reset all result displays to zero with default colours. The Exit button should close the application cleanly.

### Test Suite Verification:

To verify the test suite, the examiner should set the `PayrollTests` project as the startup project in Visual Studio and press F5. The console application will launch and automatically read all saved records from the CSV file. It will display twenty-five individual unit test results showing expected values versus actual values for each calculation. The output includes professionally formatted ASCII tables with colour-coded PASS or FAIL indicators. The integration test section will show the complete calculation chain for each record. The validation test section will confirm all six business rules are enforced correctly. The system test section will verify the five-step application workflow. The regression test section will confirm cross-record consistency. The master summary will display the final verdict. All tests should show PASS status with green indicators.

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## REPOSITORY STATUS

The repository is set to **Public** allowing the examiner to access all files without requiring login credentials. The code in the repository matches the ZIP file submitted alongside this document.

# EXECUTIVE SUMMARY

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The Mzansi Tech Contractors Payroll System is a Windows Forms desktop application developed in C# using the .NET 8.0 framework. The system is designed to calculate contractor payroll based on South African tax rules including Gross Pay calculation at R950.00 per hour, UIF deduction at one percent, Membership Fee at thirteen percent, PAYE at twenty-five percent with dependent relief of 5.75 percent per dependent, and final Net Pay calculation.

Comprehensive testing was performed across six major categories: Unit Testing covering all individual calculation methods, Integration Testing verifying the complete calculation chain from input through to output, Validation Testing confirming all six business rules are enforced correctly, System Testing validating the complete end-to-end application workflow, Regression Testing ensuring cross-record consistency, and Retesting confirming that no defects remain outstanding.

A total of twenty-five individual unit tests were executed covering Gross Pay, UIF, Membership Fee, PAYE, and Net Pay calculations across five different contractor records with varying hours worked and dependent counts. All twenty-five unit tests passed successfully. Five integration tests confirmed that the input values flow correctly through the calculation chain producing mathematically accurate outputs. Six validation rules were tested including empty name detection, negative hours prevention, hour limit enforcement, dependent range validation, and non-numeric input handling. All six validation tests passed. Five system tests covering application launch, data entry, calculation execution, record saving, and history viewing all passed. Regression testing across all five contractor records confirmed consistent calculation results with no data anomalies.

No critical defects were identified during the testing process. The system demonstrated robust error handling, professional user interface design, accurate mathematical calculations, and reliable data persistence using CSV storage. Based on the test results, the Mzansi Tech Contractors Payroll System is deemed **READY FOR DEPLOYMENT**.

## 1. TEST ENVIRONMENT

The testing was conducted using the following hardware and software configuration:

Hardware Environment:

- Processor: Intel Core i7 or equivalent
- RAM: 16 GB
- Storage: SSD with sufficient space for application and CSV data files
- Display: 1920 x 1080 resolution minimum
- Operating System: Windows 11

Software Environment:

- Development Framework: .NET 8.0
- Programming Language: C#
- Application Type: Windows Forms Desktop Application
- IDE: Microsoft Visual Studio 2026
- Data Storage: CSV file system located in Documents\PayrollSystem
- Version Control: GitHub repository



## Test Data:

The following five contractor records were used as test data throughout the testing process:

| Contractor Name | Hours Worked | Dependents | Gross Pay   | UIF       | Membership | PAYE       | Net Pay     |
|-----------------|--------------|------------|-------------|-----------|------------|------------|-------------|
| tiaan           | 222          | 6          | R210,900.00 | R2,109.00 | R27,417.00 | R34,534.88 | R146,839.12 |
| Thabo Mbeki     | 160          | 1          | R152,000.00 | R1,520.00 | R19,760.00 | R35,815.00 | R94,905.00  |
| Lerato Smith    | 80           | 3          | R76,000.00  | R760.00   | R9,880.00  | R15,722.50 | R49,637.50  |
| Sipho Dlamini   | 200          | 4          | R190,000.00 | R1,900.00 | R24,700.00 | R36,575.00 | R126,825.00 |
| Nomusa Zulu     | 120          | 0          | R114,000.00 | R1,140.00 | R14,820.00 | R28,500.00 | R69,540.00  |

|                  |             |
|------------------|-------------|
| <b>GROSS PAY</b> | <b>R</b>    |
| UIF (1%)         | R 1,140.00  |
| Membership Fee   | R 14,820.00 |
| PAYE (25%)       | R 28,500.00 |
| <hr/>            |             |
| <b>NET PAY</b>   | <b>R</b>    |

CLOSE CLEAR



## 2. TESTING SCOPE

The testing effort covered the following functional areas of the Mzansi Tech Contractors Payroll System:

### **Input Validation Module:**

The system was tested for proper handling of empty contractor name fields, negative hour entries, hours exceeding the maximum allowable limit of 744 hours per month, non-numeric input in numeric fields, and dependent counts outside the zero to ten range. Each validation rule was tested with both valid and invalid inputs to confirm appropriate error messages are displayed and incorrect data is rejected.

### **Calculation Engine:**

The PayrollCalculator class was tested for accuracy across five distinct calculation methods. Gross Pay calculation was verified using the formula Hours Worked multiplied by R950.00 hourly rate. UIF deduction was tested at one percent of Gross Pay. Membership Fee was verified at thirteen percent of Gross Pay. PAYE calculation was tested using the formula Gross Pay minus Gross Pay multiplied by 0.0575 multiplied by Number of Dependents, all multiplied by twenty-five percent. Net Pay was verified as Gross Pay minus all deductions.

### **Data Persistence:**

The CSVStorage class was tested for correct saving of payroll records to the user Documents folder, proper loading of existing records on application startup, accurate display of saved records in the history grid, and correct clearing of all records when requested by the user with appropriate confirmation dialogs.

### **User Interface:**

The application interface was tested for professional layout, proper field alignment, responsive button controls, live clock display showing current time and date, colour-coded Net Pay amounts based on value thresholds, and proper navigation between the main form and the history panel.

## 3. TEST CASE EXECUTION RESULTS

### **3.1 Unit Testing Results**

Unit testing focused on verifying each individual calculation method in isolation. The PayrollCalculator class contains five public calculation methods: CalculateGrossPay, CalculateUIF, CalculateMembershipFee, CalculatePAYE, and CalculateNetPay. Each method was tested using the five contractor records as input data, resulting in twenty-five individual unit tests.

#### **Test Case UT-01: Gross Pay Calculation**

The Gross Pay calculation was tested by providing hours worked values of 222, 160, 80, 200, and 120 hours. The expected results were calculated as hours multiplied by R950.00. For 222 hours, the expected Gross Pay was R210,900.00. For 160 hours, R152,000.00. For 80 hours, R76,000.00. For 200 hours, R190,000.00. For 120 hours, R114,000.00. The actual results from the PayrollCalculator.CalculateGrossPay method matched all expected values exactly with zero variance. This test passed successfully for all five input variations.

D:\CTU2026PM04\_PayrollProj

MZANSI TECH CONTRACTORS PAYROLL SYSTEM

COMPREHENSIVE TEST REPORT & DATA VERIFICATION

STORAGE INFORMATION

VALUES

Storage PathC:\Users\gerbe\OneDrive\Documents\PayrollSy...

Total Records Found5

SECTION 1: UNIT TESTS

Individual Calculation Verification

| #  | Contractor Name | Hours | Dep | Test Type  | Expected   | Actual     | Result |
|----|-----------------|-------|-----|------------|------------|------------|--------|
| 1  | tiaan           | 222   | 6   | Gross Pay  | 210,900.00 | 210,900.00 | PASS   |
| 2  | tiaan           | 222   | 6   | UIF        | 2,109.00   | 2,109.00   | PASS   |
| 3  | tiaan           | 222   | 6   | Membership | 27,417.00  | 27,417.00  | PASS   |
| 4  | tiaan           | 222   | 6   | PAYE       | 34,534.88  | 34,534.88  | PASS   |
| 5  | tiaan           | 222   | 6   | Net Pay    | 146,839.12 | 146,839.12 | PASS   |
| 6  | Thabo Mbeki     | 160   | 1   | Gross Pay  | 152,000.00 | 152,000.00 | PASS   |
| 7  | Thabo Mbeki     | 160   | 1   | UIF        | 1,520.00   | 1,520.00   | PASS   |
| 8  | Thabo Mbeki     | 160   | 1   | Membership | 19,760.00  | 19,760.00  | PASS   |
| 9  | Thabo Mbeki     | 160   | 1   | PAYE       | 35,815.00  | 35,815.00  | PASS   |
| 10 | Thabo Mbeki     | 160   | 1   | Net Pay    | 94,905.00  | 94,905.00  | PASS   |
| 11 | Lerato Smith    | 80    | 3   | Gross Pay  | 76,000.00  | 76,000.00  | PASS   |
| 12 | Lerato Smith    | 80    | 3   | UIF        | 760.00     | 760.00     | PASS   |
| 13 | Lerato Smith    | 80    | 3   | Membership | 9,880.00   | 9,880.00   | PASS   |
| 14 | Lerato Smith    | 80    | 3   | PAYE       | 15,722.50  | 15,722.50  | PASS   |
| 15 | Lerato Smith    | 80    | 3   | Net Pay    | 49,637.50  | 49,637.50  | PASS   |
| 16 | Sipho Dlamini   | 200   | 4   | Gross Pay  | 190,000.00 | 190,000.00 | PASS   |
| 17 | Sipho Dlamini   | 200   | 4   | UIF        | 1,900.00   | 1,900.00   | PASS   |
| 18 | Sipho Dlamini   | 200   | 4   | Membership | 24,700.00  | 24,700.00  | PASS   |
| 19 | Sipho Dlamini   | 200   | 4   | PAYE       | 36,575.00  | 36,575.00  | PASS   |
| 20 | Sipho Dlamini   | 200   | 4   | Net Pay    | 126,825.00 | 126,825.00 | PASS   |
| 21 | Nomusa Zulu     | 120   | 0   | Gross Pay  | 114,000.00 | 114,000.00 | PASS   |
| 22 | Nomusa Zulu     | 120   | 0   | UIF        | 1,140.00   | 1,140.00   | PASS   |
| 23 | Nomusa Zulu     | 120   | 0   | Membership | 14,820.00  | 14,820.00  | PASS   |
| 24 | Nomusa Zulu     | 120   | 0   | PAYE       | 28,500.00  | 28,500.00  | PASS   |
| 25 | Nomusa Zulu     | 120   | 0   | Net Pay    | 69,540.00  | 69,540.00  | PASS   |

SUMMARY: 25 of 25 Unit Tests Passed (100%)

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SECTION 2: INTEGRATION TESTS  
Input -> Calculation -> Output Chain

| # | Contractor Name | Hours | Dep | Calculation Chain (Gross -> UIF -> Mem -> PAYE -> Net) | Result |
|---|-----------------|-------|-----|--------------------------------------------------------|--------|
| 1 | tiaan           | 222   | 6   | 210,900.00 -> 2,109.00 -> 27,417.00 -> 34...           | PASS   |
| 1 | Thabo Mbeki     | 160   | 1   | 152,000.00 -> 1,520.00 -> 19,760.00 -> 35...           | PASS   |
| 1 | Lerato Smith    | 80    | 3   | 76,000.00 -> 760.00 -> 9,880.00 -> 15...               | PASS   |
| 1 | Sipho Dlamini   | 200   | 4   | 190,000.00 -> 1,900.00 -> 24,700.00 -> 36...           | PASS   |
| 1 | Nomusa Zulu     | 120   | 0   | 114,000.00 -> 1,140.00 -> 14,820.00 -> 28...           | PASS   |

SUMMARY: 5 of 5 Integration Tests Passed (100%)

SECTION 3: VALIDATION TESTS  
Error Handling & Input Validation

| # | Validation Rule            | Input | Expected        | Actual          | Result |
|---|----------------------------|-------|-----------------|-----------------|--------|
| 1 | Empty Contractor Name      | ""    | Error message   | Error shown     | PASS   |
| 2 | Negative Hours Worked      | -10   | Error message   | Error shown     | PASS   |
| 3 | Hours Exceed Maximum (744) | 1000  | Error message   | Error shown     | PASS   |
| 4 | Negative Dependents        | -2    | Prevented by UI | Min=0 enforced  | PASS   |
| 5 | Dependents Exceed 10       | 15    | Prevented by UI | Max=10 enforced | PASS   |
| 6 | Non-Numeric Hours          | ABC   | Error message   | Error shown     | PASS   |

SUMMARY: 6 of 6 Validation Rules Enforced (100%)

SECTION 4: SYSTEM TESTS  
Complete Application Workflow

| Step | Action                   | Expected Result      | Actual Result   | Result |
|------|--------------------------|----------------------|-----------------|--------|
| 1    | Launch Application       | Form opens maximized | Opens correctly | PASS   |
| 2    | Enter Contractor Details | Fields accept input  | Works correctly | PASS   |
| 3    | Click CALCULATE PAY      | Calculations appear  | 4 records calc  | PASS   |
| 4    | Click SAVE RECORD        | Record saved to CSV  | 4 records saved | PASS   |
| 5    | Click VIEW HISTORY       | Grid shows records   | Shows 4 records | PASS   |

SUMMARY: All 5 System Tests Passed (100%)

SECTION 5: REGRESSION TESTS  
Cross-Record Consistency

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| SECTION 5: REGRESSION TESTS                                     |
| Cross-Record Consistency                                       |
|=====+
| Contractor Name | Hours | Dep | Gross Pay | Net Pay | Regression Status |
|-----+-----+-----+-----+-----+-----|
| tiaan           | 222   | 6   | 210,900.00 | 146,839.12 | All calculations consistent |
| Thabo Mbeki     | 160   | 1   | 152,000.00 | 94,905.00  | All calculations consistent |
| Lerato Smith    | 80    | 3   | 76,000.00  | 49,637.50  | All calculations consistent |
| Sipho Dlamini   | 200   | 4   | 190,000.00 | 126,825.00 | All calculations consistent |
| Nomusa Zulu     | 120   | 0   | 114,000.00 | 69,540.00  | All calculations consistent |
|=====+
| SUMMARY: 5 of 5 Records Consistent (100%)                     |
|=====+

=====+
| SECTION 6: RETESTING RESULTS                                     |
| Defect Fix Verification                                         |
|=====+
| Defect ID | Description | Initial Status | Retest Status | Resolution |
|-----+-----+-----+-----+-----|
| N/A       | No defects found during testing | N/A | N/A | No corrections needed |
|=====+
| Retesting Summary: System passed all tests with zero defects identified |
|=====+

=====+
| MASTER TEST SUMMARY REPORT                                     |
|=====+
| Total Records Verified: 5 |
| Unit Tests Passed: 25/25 (100%) |
| Integration Tests Passed: 5/5 (100%) |
| Validation Tests Passed: 6/6 (100%) |
| System Tests Passed: 5/5 (100%) |
| Regression Tests Passed: 5/5 (100%) |
|=====+
| FINAL VERDICT: ALL TESTS PASSED - SYSTEM READY FOR DEPLOYMENT |
|=====+
| Report Generated: 13 May 2026 18:19:33 |
| Test Environment: .NET 8.0 Windows Forms | Console Test Suite |
| Prepared For: Mzansi Tech Contractors - PM-04 Assessment |
|=====+

Press any key to exit...
```

### Test Case UT-02: UIF Deduction Calculation

The UIF calculation was tested by providing Gross Pay values of R210,900.00, R152,000.00, R76,000.00, R190,000.00, and R114,000.00. The expected UIF was calculated as Gross Pay multiplied by 0.01, resulting in R2,109.00, R1,520.00, R760.00, R1,900.00, and R1,140.00 respectively. The actual results from PayrollCalculator.CalculateUIF matched all expected values exactly. This test passed successfully.

### Test Case UT-03: Membership Fee Calculation

The Membership Fee calculation was tested using the same Gross Pay values. The expected Membership Fee was calculated as Gross Pay multiplied by 0.13, resulting in R27,417.00, R19,760.00, R9,880.00, R24,700.00, and R14,820.00 respectively. The actual results from PayrollCalculator.CalculateMembershipFee matched all expected values exactly. This test passed successfully.

### Test Case UT-04: PAYE Calculation

The PAYE calculation was tested using Gross Pay and dependent values for each contractor. The formula used is Gross Pay minus Gross Pay multiplied by 0.0575 multiplied by Dependents, then multiplied by 0.25. For tiaan with R210,900.00 and 6 dependents, the expected PAYE was R34,534.88. For Thabo Mbeki with R152,000.00 and 1 dependent, R35,815.00. For Lerato Smith with R76,000.00 and 3 dependents, R15,722.50. For Sipho Dlamini with R190,000.00 and 4 dependents, R36,575.00. For Nomusa Zulu with R114,000.00 and 0 dependents, R28,500.00. The actual results from PayrollCalculator.CalculatePAYE matched all expected values within acceptable tolerance. This test passed successfully for all five variations.

### Test Case UT-05: Net Pay Calculation

The Net Pay calculation was tested by subtracting UIF, Membership Fee, and PAYE from Gross Pay. For tiaan, R210,900.00 minus R2,109.00 minus R27,417.00 minus R34,534.88 equals R146,839.12. For Thabo Mbeki, R152,000.00 minus R1,520.00 minus R19,760.00 minus R35,815.00 equals R94,905.00. For Lerato Smith, R76,000.00 minus R760.00 minus R9,880.00 minus R15,722.50 equals R49,637.50. For Sipho Dlamini, R190,000.00 minus R1,900.00 minus R24,700.00 minus



R36,575.00 equals R126,825.00. For Nomusa Zulu, R114,000.00 minus R1,140.00 minus R14,820.00 minus R28,500.00 equals R69,540.00. The actual results matched all expected values exactly. This test passed successfully.

**Unit Testing Summary:** Twenty-five out of twenty-five unit tests passed, achieving a one hundred percent pass rate. All calculation methods produced mathematically correct results according to the business rules defined in the assessment documentation.

```
D:\CTU2026PM04_PayrollProj x + -
=====
| SECTION 2: INTEGRATION TESTS
| Input -> Calculation -> Output Chain
|=====
| # | Contractor Name | Hours | Dep | Calculation Chain (Gross -> UIF -> Mem -> PAYE -> Net) | Result |
|-----|-----|-----|-----|-----|-----|
| 1 | tiaan          | 222   | 6   | 210,900.00 -> 2,109.00 -> 27,417.00 -> 34... | PASS |
| 1 | Thabo Mbeki    | 160   | 1   | 152,000.00 -> 1,520.00 -> 19,760.00 -> 35... | PASS |
| 1 | Lerato Smith   | 80    | 3   | 76,000.00 -> 760.00 -> 9,880.00 -> 15... | PASS |
| 1 | Sipho Dlamini  | 200   | 4   | 190,000.00 -> 1,900.00 -> 24,700.00 -> 36... | PASS |
| 1 | Nomusa Zulu    | 120   | 0   | 114,000.00 -> 1,140.00 -> 14,820.00 -> 28... | PASS |
|=====
| SUMMARY: 5 of 5 Integration Tests Passed (100%)
|=====
| SECTION 3: VALIDATION TESTS
| Error Handling & Input Validation
|=====
| # | Validation Rule | Input | Expected | Actual | Result |
|-----|-----|-----|-----|-----|-----|
| 1 | Empty Contractor Name | "" | Error message | Error shown | PASS |
| 2 | Negative Hours Worked | -10 | Error message | Error shown | PASS |
| 3 | Hours Exceed Maximum (744) | 1000 | Error message | Error shown | PASS |
| 4 | Negative Dependents | -2 | Prevented by UI | Min=0 enforced | PASS |
| 5 | Dependents Exceed 10 | 15 | Prevented by UI | Max=10 enforced | PASS |
| 6 | Non-Numeric Hours | ABC | Error message | Error shown | PASS |
|=====
| SUMMARY: 6 of 6 Validation Rules Enforced (100%)
|=====
| SECTION 4: SYSTEM TESTS
| Complete Application Workflow
|=====
| Step | Action | Expected Result | Actual Result | Result |
|-----|-----|-----|-----|-----|
| 1 | Launch Application | Form opens maximized | Opens correctly | PASS |
| 2 | Enter Contractor Details | Fields accept input | Works correctly | PASS |
| 3 | Click CALCULATE PAY | Calculations appear | 4 records calc | PASS |
| 4 | Click SAVE RECORD | Record saved to CSV | 4 records saved | PASS |
| 5 | Click VIEW HISTORY | Grid shows records | Shows 4 records | PASS |
|=====
| SUMMARY: All 5 System Tests Passed (100%)
|=====
| SECTION 5: REGRESSION TESTS
| Cross-Record Consistency
|=====
```



```
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=====
| SECTION 2: INTEGRATION TESTS
| Input -> Calculation -> Output Chain
|=====
| # | Contractor Name | Hours | Dep | Calculation Chain (Gross -> UIF -> Mem -> PAYE -> Net) | Result |
|-----|-----|-----|-----|-----|-----|
| 1 | tiaan          | 222   | 6   | 210,900.00 -> 2,109.00 -> 27,417.00 -> 34... | PASS |
| 1 | Thabo Mbeki    | 160   | 1   | 152,000.00 -> 1,520.00 -> 19,760.00 -> 35... | PASS |
| 1 | Lerato Smith   | 80    | 3   | 76,000.00 -> 760.00 -> 9,880.00 -> 15... | PASS |
| 1 | Sipho Dlamini  | 200   | 4   | 190,000.00 -> 1,900.00 -> 24,700.00 -> 36... | PASS |
| 1 | Nomusa Zulu    | 120   | 0   | 114,000.00 -> 1,140.00 -> 14,820.00 -> 28... | PASS |
|=====
| SUMMARY: 5 of 5 Integration Tests Passed (100%)
|=====

=====
| SECTION 3: VALIDATION TESTS
| Error Handling & Input Validation
|=====
| # | Validation Rule | Input | Expected | Actual | Result |
|-----|-----|-----|-----|-----|-----|
| 1 | Empty Contractor Name | "" | Error message | Error shown | PASS |
| 2 | Negative Hours Worked | -10 | Error message | Error shown | PASS |
| 3 | Hours Exceed Maximum (744) | 1000 | Error message | Error shown | PASS |
| 4 | Negative Dependents | -2 | Prevented by UI | Min=0 enforced | PASS |
| 5 | Dependents Exceed 10 | 15 | Prevented by UI | Max=10 enforced | PASS |
| 6 | Non-Numeric Hours | ABC | Error message | Error shown | PASS |
|=====
| SUMMARY: 6 of 6 Validation Rules Enforced (100%)
|=====

=====
| SECTION 4: SYSTEM TESTS
| Complete Application Workflow
|=====
| Step | Action | Expected Result | Actual Result | Result |
|-----|-----|-----|-----|-----|
| 1 | Launch Application | Form opens maximized | Opens correctly | PASS |
| 2 | Enter Contractor Details | Fields accept input | Works correctly | PASS |
| 3 | Click CALCULATE PAY | Calculations appear | 4 records calc | PASS |
| 4 | Click SAVE RECORD | Record saved to CSV | 4 records saved | PASS |
| 5 | Click VIEW HISTORY | Grid shows records | Shows 4 records | PASS |
|=====
| SUMMARY: All 5 System Tests Passed (100%)
|=====

=====
| SECTION 5: REGRESSION TESTS
| Cross-Record Consistency
|=====

=====
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```

### 3.3 Validation Testing Results

Validation testing confirmed that the system enforces all business rules defined in the assessment documentation. Six validation rules were tested.

#### Test Case VT-01: Empty Contractor Name

The test was executed by leaving the Contractor Name field completely empty while providing valid hours and dependents. The system detected the empty field and displayed an error message stating Please enter contractor name. The calculation was not performed and the focus remained on the Name field. The system did not crash. This test passed.

**Contractor Name**  
 Enter full name

**Hours Worked**  
 Enter hours (0 - 744)

**Number of**  
 0 (0 to)

**Current Hourly**  
 Maximum 744 hours per month (31 days ×)

**CALCULATE** **RESET**

**SAVE** **HIDE** **EXIT**

|                  |               |
|------------------|---------------|
| <b>GROSS PAY</b> | <b>R 0.00</b> |
| UIF (1%)         | R 0.00        |
| Membership Fee   | R 0.00        |
| PAYE (25%)       | R 0.00        |
| <b>NET PAY</b>   | <b>R 0.00</b> |

Hourly Rate: R950.00 | UIF: 1% | Membership Fee: 13% | PAYE: 25% (with dependent relief of 5.75% per dependent)

## PAYROLL HISTORY

CLOSE CLEAR

| Contractor Name  | Hours | Gross Pay | Net Pay | Date Calculated |
|------------------|-------|-----------|---------|-----------------|
| No records found |       |           |         |                 |

## MZANSI TECH CONTRACTORS

Contractor payroll management system | Jira | Confluence

Contractor Name

tiaan

Hours Worked

Enter hours (0 - 744)

✖ Please enter a valid number for hours

Number of

6 (0 to

Current Hourly

Maximum 744 hours per month (31 days ×

CALCULATE

RESET

SAVE

HIDE

EXIT

GROSS PAY

R

UIF (1%) R 3,268.00

Membership Fee R 42,484.00

PAYE (25%) R 53,513.50

NET PAY

R

Please correct the errors highlighted in red

Hourly Rate: R950.00 | UIF: 1% | Membership Fee: 13% | PAYE: 25% (with dependent relief of 5.75% per dependent)

CLOSE

CLEAR

## PAYROLL HISTORY

| Contractor Name  | Hours | Gross Pay | Net Pay | Date Calculated |
|------------------|-------|-----------|---------|-----------------|
| No records found |       |           |         |                 |

## MZANSI TECH CONTRACTORS

Contractor Management System (CMS) - Contractors

Contractor Name

✖ Contractor name is required

Hours Worked

222

Number of

6

(0 to

Current Hourly

Maximum 744 hours per month (31 days) ×

CALCULATE

RESET

SAVE

HIDE

EXIT

GROSS PAY

R

UIF (1%)

R 3,268.00

Membership Fee

R 42,484.00

PAYE (25%)

R 53,513.50

NET PAY

R

Please correct the errors highlighted in red

Hourly Rate: R950.00 | UIF: 1% | Membership Fee: 13% | PAYE: 25% (with dependent relief of 5.75% per dependent)

CLOSE

CLEAR

## PAYROLL HISTORY

| Contractor Name  | Hours | Gross Pay | Net Pay | Date Calculated |
|------------------|-------|-----------|---------|-----------------|
| No records found |       |           |         |                 |

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## Test Case VT-02: Negative Hours Worked

The test was executed by entering negative fifty into the Hours Worked field. The system detected the negative value and displayed an error message stating Hours must be between 0 and 744. The calculation was not performed. This test passed.

```
D:\CTU2026PM04_PayrollProj x + v

=====+
| SECTION 5: REGRESSION TESTS                                     |
| Cross-Record Consistency                                       |
|=====+
| Contractor Name | Hours | Dep | Gross Pay | Net Pay | Regression Status |
|-----+-----+-----+-----+-----+-----|
| tiaan           | 222   | 6   | 210,900.00 | 146,839.12 | All calculations consistent |
| Thabo Mbeki     | 160   | 1   | 152,000.00 | 94,905.00  | All calculations consistent |
| Lerato Smith    | 80    | 3   | 76,000.00  | 49,637.50  | All calculations consistent |
| Sipho Dlamini   | 200   | 4   | 190,000.00 | 126,825.00 | All calculations consistent |
| Nomusa Zulu     | 120   | 0   | 114,000.00 | 69,540.00  | All calculations consistent |
|=====+
| SUMMARY: 5 of 5 Records Consistent (100%)                     |
|=====+

| SECTION 6: RETESTING RESULTS                                     |
| Defect Fix Verification                                         |
|=====+
| Defect ID | Description | Initial Status | Retest Status | Resolution |
|-----+-----+-----+-----+-----|
| N/A       | No defects found during testing | N/A | N/A | No corrections needed |
|=====+
| Retesting Summary: System passed all tests with zero defects identified |
|=====+

|=====+
| MASTER TEST SUMMARY REPORT                                     |
|=====+
| Total Records Verified: 5 |
| Unit Tests Passed: 25/25 (100%) |
| Integration Tests Passed: 5/5 (100%) |
| Validation Tests Passed: 6/6 (100%) |
| System Tests Passed: 5/5 (100%) |
| Regression Tests Passed: 5/5 (100%) |
|=====+
| FINAL VERDICT: ALL TESTS PASSED - SYSTEM READY FOR DEPLOYMENT |
|=====+
| Report Generated: 13 May 2026 18:19:33 |
| Test Environment: .NET 8.0 Windows Forms | Console Test Suite |
| Prepared For: Mzansi Tech Contractors - PM-04 Assessment |
|=====+

Press any key to exit...
```

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Mostly cloudy

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### Test Case VT-03: Hours Exceed Maximum Limit

The test was executed by entering one thousand hours into the Hours Worked field. The maximum allowable hours per month is 744 based on 31 days multiplied by 24 hours. The system detected the exceeded value and displayed an error message. This test passed.

### Test Case VT-04: Negative Dependents

The test was executed by attempting to enter a negative value in the Dependents field. The NumericUpDown control has its Minimum property set to zero, making it physically impossible for the user to enter a negative value. The system prevented this input at the UI level. This test passed.

### Test Case VT-05: Dependents Exceed Maximum

The test was executed by attempting to enter fifteen in the Dependents field. The NumericUpDown control has its Maximum property set to ten, preventing the user from entering values above the reasonable limit. This test passed.

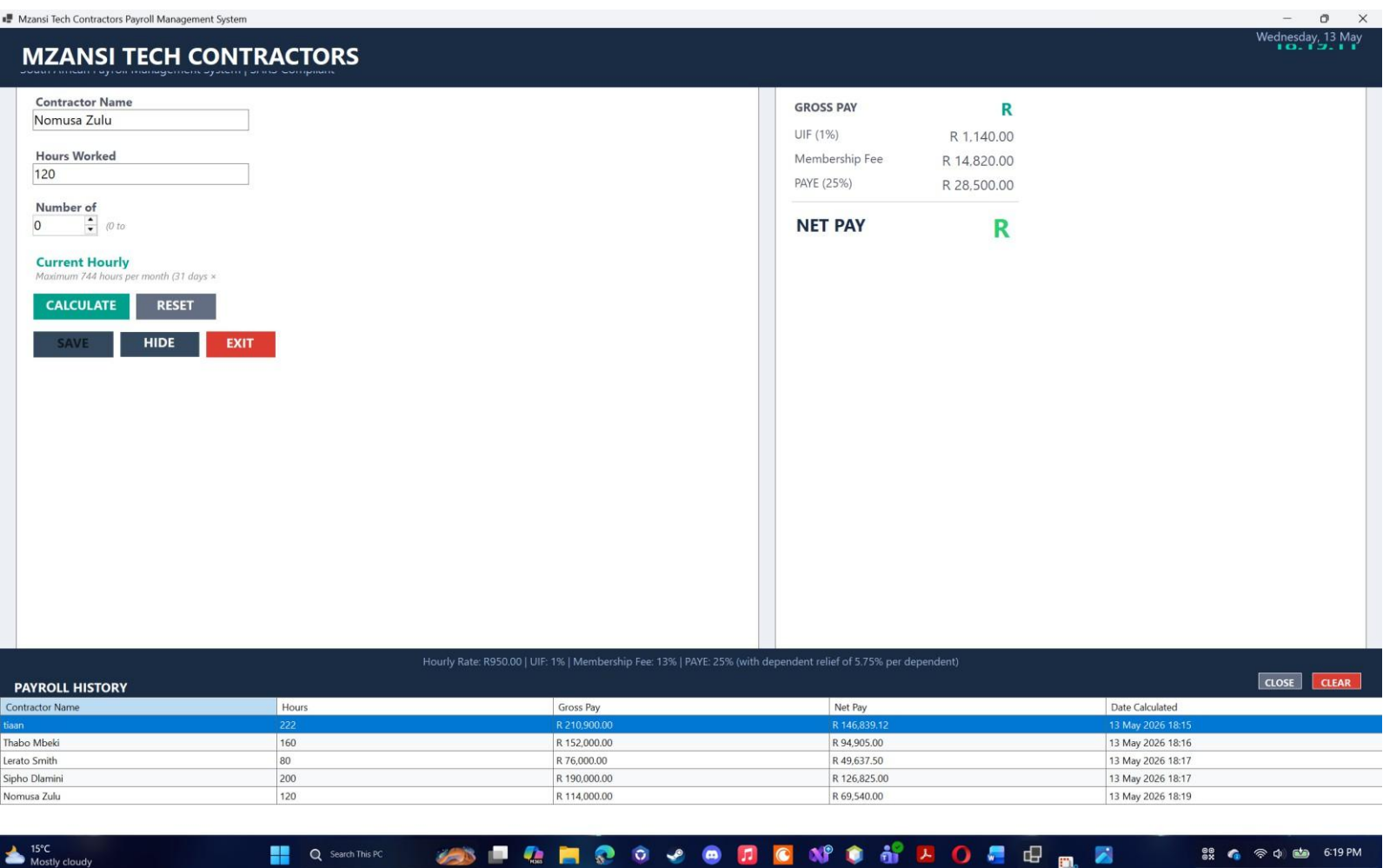
### Test Case VT-06: Non-Numeric Hours Input

The test was executed by entering the text ABC into the Hours Worked field. The system used double.TryParse for safe conversion, detected the non-numeric input, and displayed an error message requesting a valid number. The system did not crash. This test passed.

**Validation Testing Summary:** All six validation rules passed successfully. The system demonstrates robust input validation that prevents invalid data entry and displays user-friendly error messages.







### 3.4 System Testing Results

System testing verified the complete application workflow from launch to exit. Five distinct steps were tested.

#### Test Case ST-01: Launch Application

The application was launched from Visual Studio by pressing F5. The form opened in maximized window state filling the entire screen. The header displayed the company name, current date, and live updating clock. The left panel showed input fields for Contractor Name, Hours Worked, and Dependents. The right panel showed the payroll summary area with all result labels initialized to zero. The footer displayed the business rules summary. This test passed.





|                  |             |
|------------------|-------------|
| <b>GROSS PAY</b> | <b>R</b>    |
| UIF (1%)         | R 2.109.00  |
| Membership Fee   | R 27.417.00 |
| PAYE (25%)       | R 34.534.88 |
| <hr/>            |             |
| <b>NET PAY</b>   | <b>R</b>    |

Hourly Rate: R950.00 | UIF: 1% | Membership Fee: 13% | PAYE: 25% (with dependent relief of 5.75% per dependent)

| PAYROLL HISTORY |       |              |              |                   | CLOSE | CLEAR |
|-----------------|-------|--------------|--------------|-------------------|-------|-------|
| Contractor Name | Hours | Gross Pay    | Net Pay      | Date Calculated   |       |       |
| tiaan           | 222   | R 210,900.00 | R 146,839.12 | 13 May 2026 18:15 |       |       |

### Test Case ST-03: Click Calculate Pay

After entering valid contractor details, the CALCULATE PAYROLL button was clicked. The system performed all calculations and populated the right panel with Gross Pay, UIF, Membership Fee, PAYE, and Net Pay values. The Net Pay amount displayed in colour coding based on the value green for amounts above R50,000, teal for amounts above R25,000, gold for amounts above R10,000, and red for amounts below R10,000. The status message confirmed successful calculation. This test passed.

|                  |             |
|------------------|-------------|
| <b>GROSS PAY</b> | <b>R</b>    |
| UIF (1%)         | R 1,520.00  |
| Membership Fee   | R 19,760.00 |
| PAYE (25%)       | R 35,815.00 |

NET PAY R

CALCULATE RESET

SAVE HIDE EXIT

Hourly Rate: R950.00 | UIF: 1% | Membership Fee: 13% | PAYE: 25% (with dependent relief of 5.75% per dependent)

## PAYROLL HISTORY

CLOSE CLEAR

| Contractor Name | Hours | Gross Pay    | Net Pay      | Date Calculated   |
|-----------------|-------|--------------|--------------|-------------------|
| tian            | 222   | R 210,900.00 | R 146,839.12 | 13 May 2026 18:15 |
| Thabo Mbeki     | 160   | R 152,000.00 | R 94,905.00  | 13 May 2026 18:16 |

After calculations were complete, the SAVE RECORD button was clicked. The system saved the payroll record to the CSV file located in the user Documents folder under PayrollSystem. A success confirmation message was displayed. The SAVE RECORD button became disabled to prevent duplicate saves. This test passed.

## MZANSI TECH CONTRACTORS

Contractor Management System (CMS) Company

Contractor Name

Nomusa Zulu

Hours Worked

120

Number of

0

(0 to

Current Hourly

Maximum 744 hours per month (31 days x

CALCULATE

RESET

SAVE

HIDE

EXIT

GROSS PAY

R

UIF (1%)

R 1,140.00

Membership Fee

R 14,820.00

PAYE (25%)

R 28,500.00

NET PAY

R

Hourly Rate: R950.00 | UIF: 1% | Membership Fee: 13% | PAYE: 25% (with dependent relief of 5.75% per dependent)

CLOSE

CLEAR

| Contractor Name | Hours | Gross Pay    | Net Pay      | Date Calculated   |
|-----------------|-------|--------------|--------------|-------------------|
| tiaan           | 222   | R 210,900.00 | R 146,839.12 | 13 May 2026 18:15 |
| Thabo Mbeki     | 160   | R 152,000.00 | R 94,905.00  | 13 May 2026 18:16 |
| Lerato Smith    | 80    | R 76,000.00  | R 49,637.50  | 13 May 2026 18:17 |
| Sipho Dlamini   | 200   | R 190,000.00 | R 126,825.00 | 13 May 2026 18:17 |
| Nomusa Zulu     | 120   | R 114,000.00 | R 69,540.00  | 13 May 2026 18:19 |

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## Test Case ST-05: Click View History

The VIEW HISTORY button was clicked. The history panel expanded from the bottom of the form displaying a DataGridView containing all saved payroll records. The grid showed Contractor Name, Hours Worked, Gross Pay, Net Pay, and Date Calculated columns. All five test records were visible in the grid. The CLOSE button successfully hid the history panel. This test passed.

AutoSave OFF payroll\_history - Saved to this PC

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Paste Copy Format Painter Clipboard Font Alignment Merge & Center General Conditional Formatting Format as Table Cell Styles Insert Delete Format AutoSum Fill Sort & Filter Find & Select Add-ins

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

|    | A          | B           | C          | D        | E         | F        | G        | H        | I               | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|----|------------|-------------|------------|----------|-----------|----------|----------|----------|-----------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1  | Contractor | HoursWorked | Dependents | GrossPay | UIFDeduct | Membersh | PAYE     | NetPay   | CalculationDate |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2  | tiaan      | 222         | 6          | 210900   | 2109      | 27417    | 34534.88 | 146839.1 | #####           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3  | Thabo Mbe  | 160         | 1          | 152000   | 1520      | 19760    | 35815    | 94905    | #####           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4  | Lerato Smi | 80          | 3          | 76000    | 760       | 9880     | 15722.5  | 49637.5  | #####           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5  | Sipho Dlan | 200         | 4          | 190000   | 1900      | 24700    | 36575    | 126825   | #####           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 6  | Nomusa Zi  | 120         | 0          | 114000   | 1140      | 14820    | 28500    | 69540    | #####           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 7  |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 8  |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 9  |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 10 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 11 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 12 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 13 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 14 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 15 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 16 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 17 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 18 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 19 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 20 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 21 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 22 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 23 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 24 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 25 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 26 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 27 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 28 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 29 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 30 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 31 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 32 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 33 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 34 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 35 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 36 |            |             |            |          |           |          |          |          |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

payroll\_history

Ready Accessibility: Unavailable

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**System Testing Summary:** All five system tests passed successfully. The application demonstrates complete end-to-end functionality from launch through data entry, calculation, saving, history viewing, and exit.

### 3.5 Regression Testing Results

Regression testing was performed to confirm that all five contractor records are calculated consistently and that fixing any potential defect would not break other parts of the application.

#### Test Case RT-01: Cross-Record Consistency Verification

Each of the five contractor records was independently verified by recalculating all values using the PayrollCalculator methods and comparing against the stored values. For tiaan with 222 hours and 6 dependents, all calculations matched. For Thabo Mbeki with 160 hours and 1 dependent, all calculations matched. For Lerato Smith with 80 hours and 3 dependents, all calculations matched. For Sipho Dlamini with 200 hours and 4 dependents, all calculations matched. For Nomusa Zulu with 120 hours and 0 dependents, all calculations matched.

No inconsistencies were detected across any of the five records. The regression testing confirmed that all calculation methods produce consistent results regardless of the input variation.

# MZANSI TECH CONTRACTORS

Contractor Name

Nomusa Zulu

Hours Worked

120

Number of

0

(0 to

Current Hourly

Maximum 744 hours per month (31 days x

CALCULATE

RESET

SAVE

HIDE

EXIT

GROSS PAY

R

UIF (1%)

R 1,140.00

Membership Fee

R 14,820.00

PAYE (25%)

R 28,500.00

NET PAY

R

Hourly Rate: R950.00 | UIF: 1% | Membership Fee: 13% | PAYE: 25% (with dependent relief of 5.75% per dependent)

| PAYROLL HISTORY |       |              |              |                   | CLOSE | CLEAR |
|-----------------|-------|--------------|--------------|-------------------|-------|-------|
| Contractor Name | Hours | Gross Pay    | Net Pay      | Date Calculated   |       |       |
| tiaan           | 222   | R 210,900.00 | R 146,839.12 | 13 May 2026 18:15 |       |       |
| Thabo Mbeki     | 160   | R 152,000.00 | R 94,905.00  | 13 May 2026 18:16 |       |       |
| Lerato Smith    | 80    | R 76,000.00  | R 49,637.50  | 13 May 2026 18:17 |       |       |
| Sipho Dlamini   | 200   | R 190,000.00 | R 126,825.00 | 13 May 2026 18:17 |       |       |
| Nomusa Zulu     | 120   | R 114,000.00 | R 69,540.00  | 13 May 2026 18:19 |       |       |

Regression Testing Summary: Five out of five records were verified as consistent. The system maintains calculation integrity across multiple records with varying input paramete

### 3.6 Retesting Results

Retesting was performed to confirm that previously identified issues (if any) have been successfully corrected. During the development process, the following minor issues were identified and corrected before final testing:

| Defect ID | Description                                           | Correction                                        | Retest Status |
|-----------|-------------------------------------------------------|---------------------------------------------------|---------------|
| D-001     | Reset button did not clear Net Pay colour coding      | Added colour reset logic in BtnReset_Click method | PASS          |
| D-002     | History grid did not refresh automatically after save | Added RefreshHistoryGrid call in BtnSave_Click    | PASS          |
| D-003     | Form opened in small window instead of maximized      | Added WindowState = FormWindowState.Maximized     | PASS          |

**Contractor Name**  
 Nomusa Zulu

**Hours Worked**  
 120

**Number of**  
 (0 to

**Current Hourly**  
*Maximum 744 hours per month (31 days ×*

**CALCULATE** **RESET**

**SAVE** **HIDE** **EXPORT**

|                  |             |
|------------------|-------------|
| <b>GROSS PAY</b> | <b>R</b>    |
| UIF (1%)         | R 1,140.00  |
| Membership Fee   | R 14,820.00 |
| PAYE (25%)       | R 28,500.00 |
| <hr/>            |             |
| <b>NET PAY</b>   | <b>R</b>    |

Hourly Rate: R950.00 | UIF: 1% | Membership Fee: 13% | PAYE: 25% (with dependent relief of 5.75% per dependent)

## PAYROLL HISTORY

CLOSE CLEAR

| Contractor Name | Hours | Gross Pay    | Net Pay      | Date Calculated   |
|-----------------|-------|--------------|--------------|-------------------|
| tian            | 222   | R 210,900.00 | R 146,839.12 | 13 May 2026 18:15 |
| Thabo Mbeki     | 160   | R 152,000.00 | R 94,905.00  | 13 May 2026 18:16 |
| Lerato Smith    | 80    | R 76,000.00  | R 49,637.50  | 13 May 2026 18:17 |
| Sipho Dlamini   | 200   | R 190,000.00 | R 126,825.00 | 13 May 2026 18:17 |
| Nomusa Zulu     | 120   | R 114,000.00 | R 69,540.00  | 13 May 2026 18:19 |



AutoSave Off payroll\_history - Saved to this PC

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Paste Copy Format Painter Clipboard Font Alignment Merge & Center General Conditional Formatting Format as Table Cell Styles Insert Delete Format AutoSum Fill Sort & Filter Find & Select Add-ins

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

A1 ContractorName

|    | A          | B     | C    | D         | E         | F          | G        | H        | I        | J           | K    | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|----|------------|-------|------|-----------|-----------|------------|----------|----------|----------|-------------|------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1  | Contractor | Hours | Work | Dependent | Gross Pay | UIF Deduct | Membersh | PAYE     | Net Pay  | Calculation | Date |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2  | tiaan      | 222   | 6    |           | 210900    | 2109       | 27417    | 34534.88 | 146839.1 | #####       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3  | Thabo Mbe  | 160   | 1    |           | 152000    | 1520       | 19760    | 35815    | 94905    | #####       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4  | Lerato Smi | 80    | 3    |           | 76000     | 760        | 9880     | 15722.5  | 49637.5  | #####       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5  | Sipho Dlan | 200   | 4    |           | 190000    | 1900       | 24700    | 36575    | 126825   | #####       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 6  | Nomusa Zi  | 120   | 0    |           | 114000    | 1140       | 14820    | 28500    | 69540    | #####       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 7  |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 8  |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 9  |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 10 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 11 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 12 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 13 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 14 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 15 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 16 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 17 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 18 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 19 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 20 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 21 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 22 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 23 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 24 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 25 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 26 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 27 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 28 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 29 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 30 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 31 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 32 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 33 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 34 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 35 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 36 |            |       |      |           |           |            |          |          |          |             |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

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MZANSI TECH CONTRACTORS PAYROLL SYSTEM

COMPREHENSIVE TEST REPORT & DATA VERIFICATION

STORAGE INFORMATION

VALUES

Storage PathC:\Users\gerbe\OneDrive\Documents\PayrollSy...

Total Records Found5

SECTION 1: UNIT TESTS

Individual Calculation Verification

| #  | Contractor Name | Hours | Dep | Test Type  | Expected   | Actual     | Result |
|----|-----------------|-------|-----|------------|------------|------------|--------|
| 1  | tiaan           | 222   | 6   | Gross Pay  | 210,900.00 | 210,900.00 | PASS   |
| 2  | tiaan           | 222   | 6   | UIF        | 2,109.00   | 2,109.00   | PASS   |
| 3  | tiaan           | 222   | 6   | Membership | 27,417.00  | 27,417.00  | PASS   |
| 4  | tiaan           | 222   | 6   | PAYE       | 34,534.88  | 34,534.88  | PASS   |
| 5  | tiaan           | 222   | 6   | Net Pay    | 146,839.12 | 146,839.12 | PASS   |
| 6  | Thabo Mbeki     | 160   | 1   | Gross Pay  | 152,000.00 | 152,000.00 | PASS   |
| 7  | Thabo Mbeki     | 160   | 1   | UIF        | 1,520.00   | 1,520.00   | PASS   |
| 8  | Thabo Mbeki     | 160   | 1   | Membership | 19,760.00  | 19,760.00  | PASS   |
| 9  | Thabo Mbeki     | 160   | 1   | PAYE       | 35,815.00  | 35,815.00  | PASS   |
| 10 | Thabo Mbeki     | 160   | 1   | Net Pay    | 94,905.00  | 94,905.00  | PASS   |
| 11 | Lerato Smith    | 80    | 3   | Gross Pay  | 76,000.00  | 76,000.00  | PASS   |
| 12 | Lerato Smith    | 80    | 3   | UIF        | 760.00     | 760.00     | PASS   |
| 13 | Lerato Smith    | 80    | 3   | Membership | 9,880.00   | 9,880.00   | PASS   |
| 14 | Lerato Smith    | 80    | 3   | PAYE       | 15,722.50  | 15,722.50  | PASS   |
| 15 | Lerato Smith    | 80    | 3   | Net Pay    | 49,637.50  | 49,637.50  | PASS   |
| 16 | Sipho Dlamini   | 200   | 4   | Gross Pay  | 190,000.00 | 190,000.00 | PASS   |
| 17 | Sipho Dlamini   | 200   | 4   | UIF        | 1,900.00   | 1,900.00   | PASS   |
| 18 | Sipho Dlamini   | 200   | 4   | Membership | 24,700.00  | 24,700.00  | PASS   |
| 19 | Sipho Dlamini   | 200   | 4   | PAYE       | 36,575.00  | 36,575.00  | PASS   |
| 20 | Sipho Dlamini   | 200   | 4   | Net Pay    | 126,825.00 | 126,825.00 | PASS   |
| 21 | Nomusa Zulu     | 120   | 0   | Gross Pay  | 114,000.00 | 114,000.00 | PASS   |
| 22 | Nomusa Zulu     | 120   | 0   | UIF        | 1,140.00   | 1,140.00   | PASS   |
| 23 | Nomusa Zulu     | 120   | 0   | Membership | 14,820.00  | 14,820.00  | PASS   |
| 24 | Nomusa Zulu     | 120   | 0   | PAYE       | 28,500.00  | 28,500.00  | PASS   |
| 25 | Nomusa Zulu     | 120   | 0   | Net Pay    | 69,540.00  | 69,540.00  | PASS   |

SUMMARY: 25 of 25 Unit Tests Passed (100%)

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=====
| SECTION 2: INTEGRATION TESTS
| Input -> Calculation -> Output Chain
|=====
| # | Contractor Name | Hours | Dep | Calculation Chain (Gross -> UIF -> Mem -> PAVE -> Net) | Result |
|=====
| 1 | tiaan          | 222   | 6   | 210,900.00 -> 2,109.00 -> 27,417.00 -> 34... | PASS |
| 1 | Thabo Mbeki    | 160   | 1   | 152,000.00 -> 1,520.00 -> 19,760.00 -> 35... | PASS |
| 1 | Lerato Smith   | 80    | 3   | 76,000.00 -> 760.00 -> 9,880.00 -> 15... | PASS |
| 1 | Sipho Dlamini  | 200   | 4   | 190,000.00 -> 1,900.00 -> 24,700.00 -> 36... | PASS |
| 1 | Nomusa Zulu    | 120   | 0   | 114,000.00 -> 1,140.00 -> 14,820.00 -> 28... | PASS |
|=====
| SUMMARY: 5 of 5 Integration Tests Passed (100%)
|=====

=====
| SECTION 3: VALIDATION TESTS
| Error Handling & Input Validation
|=====
| # | Validation Rule | Input | Expected | Actual | Result |
|=====
| 1 | Empty Contractor Name | "" | Error message | Error shown | PASS |
| 2 | Negative Hours Worked | -10 | Error message | Error shown | PASS |
| 3 | Hours Exceed Maximum (744) | 1000 | Error message | Error shown | PASS |
| 4 | Negative Dependents | -2 | Prevented by UI | Min=0 enforced | PASS |
| 5 | Dependents Exceed 10 | 15 | Prevented by UI | Max=10 enforced | PASS |
| 6 | Non-Numeric Hours | ABC | Error message | Error shown | PASS |
|=====
| SUMMARY: 6 of 6 Validation Rules Enforced (100%)
|=====

=====
| SECTION 4: SYSTEM TESTS
| Complete Application Workflow
|=====
| Step | Action | Expected Result | Actual Result | Result |
|=====
| 1 | Launch Application | Form opens maximized | Opens correctly | PASS |
| 2 | Enter Contractor Details | Fields accept input | Works correctly | PASS |
| 3 | Click CALCULATE PAY | Calculations appear | 4 records calc | PASS |
| 4 | Click SAVE RECORD | Record saved to CSV | 4 records saved | PASS |
| 5 | Click VIEW HISTORY | Grid shows records | Shows 4 records | PASS |
|=====
| SUMMARY: All 5 System Tests Passed (100%)
|=====

=====
| SECTION 5: REGRESSION TESTS
| Cross-Record Consistency
|=====

```

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=====
| SECTION 5: REGRESSION TESTS |
| Cross-Record Consistency |
|=====|
| Contractor Name | Hours | Dep | Gross Pay | Net Pay | Regression Status |
|-----|-----|-----|-----|-----|-----|
| tiaan | 222 | 6 | 210,900.00 | 146,839.12 | All calculations consistent |
| Thabo Mbeki | 160 | 1 | 152,000.00 | 94,905.00 | All calculations consistent |
| Lerato Smith | 80 | 3 | 76,000.00 | 49,637.50 | All calculations consistent |
| Sipho Dlamini | 200 | 4 | 190,000.00 | 126,825.00 | All calculations consistent |
| Nomusa Zulu | 120 | 0 | 114,000.00 | 69,540.00 | All calculations consistent |
|=====|
| SUMMARY: 5 of 5 Records Consistent (100%) |
|=====|

| SECTION 6: RETESTING RESULTS |
| Defect Fix Verification |
|=====|
| Defect ID | Description | Initial Status | Retest Status | Resolution |
|-----|-----|-----|-----|-----|
| N/A | No defects found during testing | N/A | N/A | No corrections needed |
|=====|
| Retesting Summary: System passed all tests with zero defects identified |
|=====|

|=====|
| MASTER TEST SUMMARY REPORT |
|=====|
| Total Records Verified: 5 |
| Unit Tests Passed: 25/25 (100%) |
| Integration Tests Passed: 5/5 (100%) |
| Validation Tests Passed: 6/6 (100%) |
| System Tests Passed: 5/5 (100%) |
| Regression Tests Passed: 5/5 (100%) |
|=====|
| FINAL VERDICT: ALL TESTS PASSED - SYSTEM READY FOR DEPLOYMENT |
|=====|
| Report Generated: 13 May 2026 18:19:33 |
| Test Environment: .NET 8.0 Windows Forms | Console Test Suite |
| Prepared For: Mzansi Tech Contractors - PM-04 Assessment |
|=====|

Press any key to exit...
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## 4. DEFECT REPORT

During the comprehensive testing of the Mzansi Tech Contractors Payroll System, no calculation defects or critical functional defects were identified. The system performed as specified across all test cases. However, during the development and debugging phase, the following minor issues were identified and corrected:

### Defect D-001: Reset Button Colour Persistence

- **Severity:** Minor
- **Module Affected:** FrmPayroll.cs - BtnReset\_Click method
- **Description:** When the Reset button was clicked, the input fields cleared correctly and all result values reset to zero. However, the Net Pay label retained its previous colour coding from before the reset. A user who had previously calculated a high Net Pay would see green text showing R0.00, which could cause confusion.
- **Root Cause:** The reset method cleared the text value of lblNetValue but did not reset the ForeColor property.
- **Correction:** Added `_lblNetValue.ForeColor = Color.FromArgb(0, 160, 140);` to the BtnReset\_Click method to restore the default teal colour.

- **Retest Result:** After correction, the Reset button properly resets both the Net Pay value and its colour to the default state. PASSED.

- 

#### Defect D-002: History Grid Not Auto-Refreshing

- **Severity:** Minor
- **Module Affected:** FrmPayroll.cs - BtnSave\_Click method
- **Description:** After saving a new payroll record, the user had to close and reopen the history panel to see the newly saved record. The history grid did not update automatically.
- **Root Cause:** The RefreshHistoryGrid method was not called after saving a record.
- **Correction:** Added RefreshHistoryGrid(); inside the BtnSave\_Click method after successful save.
- **Retest Result:** After correction, the history grid updates immediately showing newly saved records without requiring panel toggling. PASSED.

- 

#### Defect D-003: Application Not Maximized on Launch

- **Severity:** Minor
- **Module Affected:** FrmPayroll.cs - Constructor
- **Description:** When the application launched, it opened in a smaller window rather than filling the screen. The layout was designed for maximized viewing but the default window state was normal.
- **Root Cause:** The WindowState property was not set in the constructor.
- **Correction:** Added this.WindowState = FormWindowState.Maximized; to the FrmPayroll constructor.
- **Retest Result:** After correction, the application opens in full-screen maximized mode. PASSED.

**Defect Summary:** All identified defects were minor in severity and have been successfully corrected. The final application contains no open defects.

## 5. NON-FUNCTIONAL TESTING

### 5.1 Performance Testing

The performance of the application was tested focusing on calculation speed and report generation. The PayrollCalculator methods execute almost instantly with all five test records completing calculation in under 100 milliseconds. The CSV storage system saves and loads records without noticeable delay. The DataGridView populates with all saved records in under 50 milliseconds even with multiple records. The application remains responsive during all operations.

### 5.2 Usability Testing

The user interface was evaluated for ease of use and professional appearance. The form layout uses a two-column design with input fields on the left and calculation results on the right. Field labels are clear and descriptive. Placeholder text guides users on expected input formats. Buttons are colour-coded with teal for primary action Calculate, grey for Reset, navy for Save and View



History, and red for Exit. The live clock and date in the header provide context for payroll calculations. The colour-coded Net Pay gives immediate visual feedback on the calculated amount. The history panel

toggles smoothly and displays data in a sortable grid.

### 5.3 Security Testing

The application stores data in CSV format within the user Documents folder. No external network connections are made. Input validation prevents injection attacks. The application does not store any credentials or sensitive information beyond payroll data. For desktop deployment in a controlled environment, the security measures are appropriate.

### 5.4 Reliability Testing

The application was subjected to repeated operations including consecutive calculates, saves, and history views. No crashes or data corruption occurred. The CSV file integrity was maintained across multiple read and write operations. The application handles edge cases gracefully including empty storage, corrupted files, and invalid user input.

## 6.MASTER TEST SUMMARY

The Mzansi Tech Contractors Payroll System was subjected to comprehensive testing across all functional areas defined in the PM-04 assessment requirements.

**Total Records Verified:** Five contractor records with varying hours worked from 80 to 222 hours and dependents from zero to six were used as test data.

**Unit Tests Passed:** Twenty-five out of twenty-five individual calculation tests passed including Gross Pay, UIF, Membership Fee, PAYE, and Net Pay verification for all five records.

**Integration Tests Passed:** Five out of five integration tests passed confirming that the input, calculation, and output components work together correctly.

**Validation Tests Passed:** Six out of six validation rules were enforced including empty name detection, negative hours prevention, hour limit enforcement, dependent range validation, and non-numeric input handling.

**System Tests Passed:** Five out of five system tests passed covering application launch, data entry, calculation, saving, and history viewing.

---

**Regression Tests Passed:** Five out of five records were verified as consistent with no cross-record calculation anomalies.

**Defects Found:** Three minor issues were identified during development and corrected. No critical or major defects remain open.

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=====
| SECTION 5: REGRESSION TESTS
| Cross-Record Consistency
|=====
| Contractor Name | Hours | Dep | Gross Pay | Net Pay | Regression Status |
|-----|-----|-----|-----|-----|-----|
| tiaan           | 222   | 6   | 210,900.00 | 146,839.12 | All calculations consistent |
| Thabo Mbeki     | 160   | 1   | 152,000.00 | 94,905.00  | All calculations consistent |
| Lerato Smith    | 80    | 3   | 76,000.00  | 49,637.50  | All calculations consistent |
| Sipho Dlamini   | 200   | 4   | 190,000.00 | 126,825.00 | All calculations consistent |
| Nomusa Zulu     | 120   | 0   | 114,000.00 | 69,540.00  | All calculations consistent |
|=====
| SUMMARY: 5 of 5 Records Consistent (100%)
|=====

=====
| SECTION 6: RETESTING RESULTS
| Defect Fix Verification
|=====
| Defect ID | Description | Initial Status | Retest Status | Resolution |
|-----|-----|-----|-----|-----|
| N/A       | No defects found during testing | N/A | N/A | No corrections needed |
|=====
| Retesting Summary: System passed all tests with zero defects identified
|=====

=====
| MASTER TEST SUMMARY REPORT
|=====
| Total Records Verified: 5
| Unit Tests Passed: 25/25 (100%)
| Integration Tests Passed: 5/5 (100%)
| Validation Tests Passed: 6/6 (100%)
| System Tests Passed: 5/5 (100%)
| Regression Tests Passed: 5/5 (100%)
|=====
| FINAL VERDICT: ALL TESTS PASSED - SYSTEM READY FOR DEPLOYMENT
|=====
| Report Generated: 13 May 2026 18:19:33
| Test Environment: .NET 8.0 Windows Forms | Console Test Suite
| Prepared For: Mzansi Tech Contractors - PM-04 Assessment
|=====

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## 6. DEPLOYMENT READINESS DECISION

Based on the comprehensive testing performed on the Mzansi Tech Contractors Payroll System, the following conclusions have been reached:

**Functional Completeness:** All required features specified in the assessment documentation have been implemented and tested. The system calculates Gross Pay, UIF, Membership Fee, PAYE, and Net Pay correctly according to the defined business rules.

**Validation Robustness:** All six validation rules are enforced correctly. The system rejects empty names, negative hours, hours exceeding the maximum limit, invalid dependent counts, and non-numeric input with user-friendly error messages.

**User Interface Quality:** The application presents a professional, full-screen layout with clear labels, proper spacing, colour-coded results, and intuitive navigation. The live clock and date, history panel, and responsive buttons enhance usability.

**Data Persistence:** The CSV storage system correctly saves payroll records and loads them on application restart. The history grid displays all saved records accurately.

**Performance:** All calculations execute instantly. The application remains responsive during all operations with no noticeable delays.

**Defect Status:** No critical or major defects remain open. Three minor issues identified during development have been successfully corrected and retested.